

```

3
4 char incomingOption;
5
6 void setup() {
7   // put your setup code here, to run once:
8   Serial.begin(9600);
9   pinMode(5, OUTPUT); // Relay1
10  pinMode(6, OUTPUT); // Relay2
11  pinMode(7, OUTPUT); // Relay3
12  pinMode(8, OUTPUT); // Relay4
13  pinMode(9, OUTPUT); // Relay5
14  pinMode(10, OUTPUT); // Relay6
15  pinMode(11, OUTPUT); // Relay7
16  pinMode(12, OUTPUT); // Relay8
17  //*****
18  pinMode(13, OUTPUT); // LED
19  //*****
20  digitalWrite(5, LOW);
21  digitalWrite(6, LOW);
22  digitalWrite(7, LOW);
23  digitalWrite(8, LOW);
24  digitalWrite(9, LOW);
25  digitalWrite(10, LOW);
26  digitalWrite(11, LOW);
27  digitalWrite(12, LOW);
28  //*****
29 }
30 void loop() {
31   // put your main code here, to run repeatedly:
32   incomingOption = Serial.read();
33
34   //*****RELAY NUMBER-1*****
35   {

```

Fig. 4: Arduino code snippet

Software programming

Install the IDE for VB development. Ensure proper USB drivers for the microcontroller or interface module. Include libraries for USB communication. Use input-output ports for serial communication. Two programs are used:

1. Arduino code sketch for the microcontroller (Fig. 4).
2. Visual Basic-based GUI for the PC, providing a graphical interface to control the appliances via the

serial port.

Programming procedure.

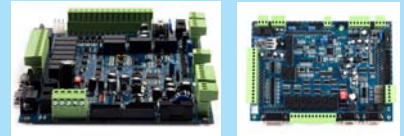
1. Assemble the circuit according to the diagram on a breadboard or zero PCB. Verify all connections.
2. If programming through Arduino Uno, the USB-to-TTL serial converter is unnecessary.
3. After programming, remove the ATmega328P IC from the Arduino Uno board and insert it into the breadboard during assembly. Add a new ATmega328P to the Arduino board.

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RS232 to RS485-422 Converter

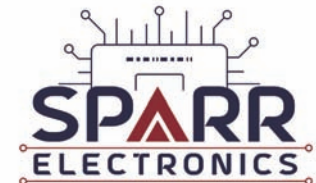
USB to RS232 Converter



RS485-422 Isolator Repeater

USB to RS485-422 Converter

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